

**Did Generative AI Reshape Employer Skill Demand?
Evidence from U.S. Federal Job Postings, 2017-2026**

Research Project Proposal

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Introduction

The rapid public diffusion of generative artificial intelligence in late 2022 marked a visible technological turning point. Large language models quickly entered professional workflows and policy discussions, raising concerns about how AI may reshape labor markets. While much of the debate has focused on employment levels and wage inequality, less attention has been paid to a more immediate signal of labor demand: employer expectations.

Job postings reflect what employers actively seek in workers. They describe required skills, competencies, and responsibilities. If generative AI altered workplace processes, such changes should be observable in the language used in job descriptions. Rather than examining employment counts or macroeconomic indicators, this project studies whether the structure of skill language in U.S. federal job postings shifted after the generative AI inflection point.

Importantly, generative AI diffusion represents a common shock across occupations. However, occupations differ in their pre-existing exposure to AI technologies. This cross-occupational variation allows us to examine whether occupations that were more technologically exposed to AI exhibited larger changes in employer skill language after late 2022.

The central research question is:

Did occupations with higher pre-existing AI exposure experience larger post-2022 shifts in skill language in federal job postings?

Theory and Prior Scholarship

A large body of research shows that technological change reshapes labor markets primarily by reorganizing tasks within occupations, rather than simply eliminating jobs. In a seminal contribution, Autor, Levy, and Murnane (2003) argue that computerization substitutes for workers performing routine tasks while complementing those engaged in nonroutine analytical and interpersonal activities. As firms adopt new technologies, routine activities are increasingly automated, while workers shift toward tasks requiring judgment, problem-solving, and complex communication. This task-based perspective suggests that technological change alters the composition of skills demanded by employers, rather than uniformly reducing labor demand.

Subsequent work expands this framework by emphasizing the multiple channels through which automation affects labor markets. Automation can displace workers in specific tasks, but it can also generate productivity gains that increase demand for complementary labor. In their task-based model of automation, Acemoglu and Restrepo (2020) highlight the interaction between displacement effects, where machines substitute for human labor, and productivity effects, where automation increases output and creates demand for new tasks. Because these mechanisms operate simultaneously, the net impact of technological change depends on how technologies interact with the tasks performed in different occupations.

A related line of research focuses on the changing importance of skill complementarities in technology-intensive workplaces. Using evidence from U.S. labor markets, Deming (2017) documents a substantial rise in demand for jobs requiring strong social and interpersonal skills, particularly in occupations that combine technical knowledge with coordination and communication. One interpretation is that technological systems increase the value of human abilities that machines struggle to replicate, such as collaboration, persuasion, and team coordination. In this sense, technological change may shift labor demand toward skills that complement new technologies rather than directly compete with them.

Recent advances in artificial intelligence extend this task-based framework to a new class of technologies capable of performing complex cognitive activities. Unlike earlier waves of automation that focused primarily on routine manual or clerical tasks, AI systems increasingly perform tasks involving pattern recognition, prediction, and decision-making. To understand how these capabilities interact with labor markets, researchers have developed measures that quantify the degree to which occupations are exposed to AI technologies. Felten, Raj, and Seamans (2021) construct an AI Occupational Exposure (AIOE) index by linking specific AI applications to occupational abilities recorded in the O*NET database. Their measure captures the extent to which the abilities required in a given occupation overlap with the capabilities of existing AI systems.

A complementary approach is proposed by Webb (2020), who measures occupational exposure to automation by analyzing the textual similarity between patent descriptions of technologies and occupational task descriptions. By matching verb–noun pairs extracted from patent texts with those appearing in job task descriptions, Webb constructs an index of how strongly a given technology targets the tasks performed within an occupation. Applying this method to artificial intelligence patents, he finds that AI exposure is concentrated in high-skill occupations involving analytical and information-processing tasks, suggesting that AI may affect a different segment of the labor market than earlier automation technologies (Webb, 2020).

Taken together, these strands of literature suggest a common mechanism. Technological change alters the task composition of occupations, which then reshapes the types of skills employers seek in workers. Occupations whose tasks overlap more strongly with emerging technological capabilities are likely to experience larger adjustments in skill requirements. If artificial intelligence represents a new wave of technological diffusion affecting cognitive and information-processing tasks, then occupations with higher exposure to AI technologies should exhibit more pronounced shifts in the types of skills emphasized in hiring.

Building on this framework, this study examines whether the diffusion of generative AI technologies is associated with measurable changes in the language used in job postings. By combining job-posting data with occupation-level measures of AI exposure, the analysis evaluates whether occupations that are more exposed to AI technologies show larger shifts in the skills and competencies employers emphasize in recruitment. This approach allows us to observe how technological change manifests directly in employer demand for labor.

Data

The analysis uses a structured dataset of U.S. federal job postings from 2017 through 2026. Each observation includes job title, occupational series code, posting date, full job description text, salary range, agency, and location.

Federal occupational series codes are mapped to six-digit SOC codes using official crosswalks. These SOC codes are then merged with the occupation-level AIOE exposure scores developed by Felten and co-authors (2021). Because the exposure index is constructed using occupational ability data and AI application mapping independent of the 2022 diffusion event, exposure is predetermined and constant over time for each occupation.

The period is divided into a pre-period (January 2017 to October 2022) and a post-period (November 2022 onward). The unit of analysis is occupation-by-time (e.g., occupation-by-month), constructed by aggregating job-level text measures.

As a robustness check, we also plan to examine industry-level job posting data. Our main dataset provides a consistent source of occupational information. However, federal hiring may respond more slowly to technological shocks due to institutional procedures and standardized job classifications. In contrast, private-sector employers are likely to adjust job requirements more quickly as technologies such as generative AI diffuse. Comparing federal postings with industry-level job posting data allows us to assess whether the observed patterns reflect broader labor market adjustments rather than features specific to federal hiring.

Research Design

The empirical strategy exploits cross-occupational variation in pre-existing AI exposure combined with the common generative AI shock in late 2022.

The baseline model

$$Y_{it} = \alpha + \beta(Post_t \times Exposure_i) + \gamma_i + \delta_t + \varepsilon_{it}$$

Where,

Y_{it} represents a measure of skill intensity or topic prevalence for occupation (i) at time (t),

$Post_t$ equals one for postings after November 2022,

$Exposure_i$ is the continuous AIOE exposure score for occupation (i),

γ_i are occupation-fixed effects,

δ_t are time-fixed effects.

SEs clustered at the occupation-series/SOC level.

Exposure is treated as continuous rather than binary, preserving the full variation in AI intensity across occupations. The interaction term captures whether the post-2022 shift in skill language varies with exposure intensity. Before 2022, the interaction equals zero for all occupations. After 2022, the magnitude of the shift depends proportionally on the exposure score.

The coefficient β measures how strongly post-2022 language changes scale with exposure. A positive and statistically significant β indicates that occupations with higher pre-existing AI exposure experienced larger increases in the outcome after 2022. Importantly, exposure itself does not change; rather, the model tests whether occupations that were more technologically aligned with AI responded more strongly to the generative AI shock.

It is not automatically the case that occupations with higher AI exposure must exhibit larger language changes after 2022. The AIOE measure reflects technological potential—how closely an occupation’s ability profile aligns with AI capabilities (Felten et al., 2021). However, potential exposure does not necessarily imply immediate adoption or organizational adjustment. Federal hiring practices may respond slowly to technological developments due to institutional constraints, regulatory structures, or budgetary considerations. In addition, language shifts in job postings could occur broadly across occupations, independent of exposure intensity, reflecting general technological awareness rather than targeted restructuring. For these reasons, whether pre-existing AI exposure translates into observable changes in employer skill language remains an empirical question rather than a mechanical outcome.

Event-study specification

We also estimate a dynamic event-study specification to examine how the effects evolve over time following the diffusion of generative AI. Rather than imposing a single post-treatment effect, this approach allows us to observe whether changes in skill requirements emerge gradually and whether they are larger in occupations with higher AI exposure. The period immediately before the diffusion of generative AI is used as the omitted reference period, so the estimated coefficients capture changes relative to the pre-shock baseline.

$$Y_{it} = \alpha + \sum_{\{k \neq -1\}} \beta_k (D_t^k \times Exposure_i) + \gamma_i + \delta_t + \varepsilon_{it}$$

Where, D_t^k denotes event time relative to the diffusion of generative AI, and $\{k \neq -1\}$, the period just before the shock, is omitted as the reference category. SEs clustered at the occupation-series/SOC level.

Identification Strategy

Identification relies on combining a common time shock with predetermined cross-occupational variation in AI exposure. The public diffusion of generative AI in November 2022 represents a nationwide technological event that affected all occupations simultaneously. However,

occupations differ in how strongly their required abilities align with AI capabilities, as measured by the AI Occupational Exposure Index (AIOE) developed by Felten, Raj, and Seamans (2021). Because this exposure index is constructed using occupational ability data and mappings to AI applications that predate the generative AI shock, it is fixed over time and not influenced by post-2022 language changes. This provides a source of exogenous variation in susceptibility to the AI shock. The empirical strategy therefore compares how skill language evolves across occupations with different exposure intensities before and after November 2022. The key identifying variation comes from whether more AI-exposed occupations exhibit larger post-2022 shifts relative to less-exposed occupations.

The credibility of this approach depends on a parallel trends assumption adapted to continuous treatment. Specifically, in the absence of generative AI diffusion, exposure intensity should not predict differential trends in skill language. In other words, occupations with higher and lower AI exposure should exhibit similar pre-2022 trends. We assess this assumption using an event-study specification that interacts exposure with event-time indicators relative to November 2022. If exposure does not predict systematic divergence prior to the shock, and differential changes emerge only after 2022, this pattern strengthens the interpretation that generative AI diffusion is associated with differential restructuring of employer expectations.

Measurement of Text-Based Outcomes

Knowledge Mining Workflow (Text Measures)

We treat job postings as text data that reveal what employers say they want. Using the description and duties fields, we clean the text (basic normalization), tokenize and lemmatize it, and remove stop words so language is comparable over time. We then apply core knowledge mining techniques from the project slides—keyword extraction, frequency comparison, TF-IDF weighting, and topic modeling (LDA) to quantify how the content of postings changes.

These text measures correspond to the project’s core knowledge mining pipeline (keyword extraction, frequency comparison, and topic modeling) used to track shifts in employer skill language over time.

Outcome measures (constructed from the workflow above)

- AI keyword intensity: frequency of AI/LLM-related terms (normalized by document length).
- Technical skill intensity: density of programming, data, and digital tool language (frequency + TF-IDF).
- Soft skill intensity: density of communication, leadership, and coordination language (frequency + TF-IDF).

- Topic prevalence: thematic emphasis estimated using LDA, summarized as topic shares over time.

These measures are aggregated to the occupation-by-time level and used as dependent variables in the DiD and event-study specifications.

Expected Contributions and Limitations

This project contributes to the literature by linking AI exposure measures to real-time employer expectations. Rather than focusing solely on wages or employment, it examines whether technological capability overlap predicts changes in hiring language following a major AI diffusion event.

Methodologically, the study demonstrates how exposure-based identification strategies can be combined with text mining techniques to study technological change. Substantively, it provides evidence on whether generative AI has begun reshaping federal employer expectations.

Limitations include the relatively slow institutional response of federal hiring and the challenge of isolating generative AI effects from broader digital trends. Nevertheless, by exploiting cross-occupational exposure variation, the analysis provides structured evidence on whether AI-aligned occupations experienced differential restructuring in skill language after 2022.

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